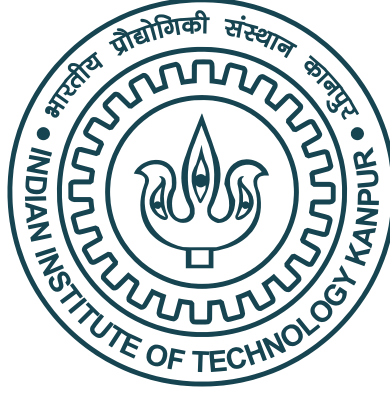




Moreau-Yosida Parallel Tempering

MTH496: Under Graduate Project - I

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Abstract

This report investigates Moreau-Yosida Parallel Tempering (MYPT), an advanced Markov Chain Monte Carlo (MCMC) sampling strategy designed to overcome severe mode entrapment in highly multimodal distributions. Standard sampling methods frequently become stuck in local modes separated by deep, low-probability valleys. While standard Parallel Tempering attempts to solve this by flattening the energy landscape at high temperatures, it suffers from theoretical integrability limitations. By integrating Moreau-Yosida envelopes, this approach mathematically smooths and convexifies the target potential energy function, ensuring a proper probability density while facilitating efficient global exploration. We analytically derive the smoothing parameters and evaluate the algorithm's performance on complex Gaussian mixtures and non-differentiable targets like the Bayesian Lasso. Quantitative evaluations confirm that the MYPT approach significantly increases the Expected Squared Jump Distance (ESJD) compared to standard methods, providing superior transition efficiency and chain mixing across challenging, disconnected landscapes.

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1 Introduction

Markov Chain Monte Carlo (MCMC) is a class of computational algorithms designed to sample from complex, high-dimensional probability distributions. It achieves this by constructing a Markov chain that converges to the target distribution, yielding correlated samples due to the dependence between consecutive states. In non-convex, multimodal distributions, it frequently traps gradient-based samplers in local minima, severely limiting state-space exploration. While Parallel Tempering (PT) solves this entrapment

by running multiple chains simultaneously at varying temperatures to flatten energy barriers, it cannot resolve the issues of extreme non-convexity or non-differentiability.

To address these challenges, this report utilizes Moreau-Yosida Envelopes to create a sequence of smoothed approximations of the target distribution. Controlled by a smoothing parameter t , where the sequence converges to the target as $t \rightarrow 0$, this technique convexifies complex geometries and smooths sharp boundaries. Integrating this convexification with the state-swapping mechanics of Parallel Tempering yields Moreau-Yosida Parallel Tempering (MYPT). The efficiency of this framework is evaluated on two cases: a severely multi-modal 20-component Gaussian mixture and the non-differentiable Bayesian Lasso.

1.1 Markov Chain Monte Carlo: Basics and Limitations

The Metropolis-Hastings algorithm generates a Markov chain by proposing a candidate state y from a current state x according to a proposal distribution $q(y|x)$. This proposed state is accepted with a probability $\alpha = \min\left(1, \frac{\pi(y)q(x|y)}{\pi(x)q(y|x)}\right)$, which ensures that the chain ultimately converges to the desired target distribution $\pi(x)$. If the proposal is rejected, the chain remains at its current state x .

A standard implementation of this framework is the Random Walk Metropolis-Hastings (RWMH) algorithm. To illustrate its methodology, consider sampling from a multi-modal target distribution comprising a four-component Gaussian mixture, using a Gaussian proposal $q(y|x) \sim \mathcal{N}(x, \sigma_p^2)$. This target density is defined as:

$$\pi(x) = \sum_{i=1}^4 w_i \phi(x|\mu_i, \sigma_i), \quad (1)$$

where $\phi(x|\mu, \sigma)$ denotes the probability density function of a normal distribution with mean μ and standard deviation σ . The weights w , means μ , and standard deviations σ for the four individual components are set to:

$$\begin{aligned} w &= (0.2, 0.2, 0.3, 0.3) \\ \mu &= (-2, -1, 1, 2) \\ \sigma &= (0.05, 0.25, 0.25, 0.1). \end{aligned}$$

Because the proposal is symmetric ($q(y|x) = q(x|y)$), the proposal density terms cancel

out, simplifying the acceptance ratio strictly to the ratio of the target densities:

$$\alpha = \min \left(1, \frac{\pi(y)}{\pi(x)} \right). \quad (2)$$

Algorithm 1 Metropolis-Hastings Algorithm

```

1: Initialize  $x^{(0)} = x_0$ 
2: for  $i = 0$  to  $N - 1$  do
3:   Draw a candidate state  $y \sim q(y|x^{(i)})$ 
4:   Calculate the acceptance probability:

```

$$\alpha = \min \left(1, \frac{\pi(y)q(x^{(i)}|y)}{\pi(x^{(i)})q(y|x^{(i)})} \right)$$

```

5:   Draw a random value  $u \sim \text{Uniform}(0, 1)$ 
6:   if  $u < \alpha$  then
7:     Accept the proposal:  $x^{(i+1)} = y$ 
8:   else
9:     Reject the proposal:  $x^{(i+1)} = x^{(i)}$ 
10:  end if
11: end for
12: return Samples  $\{x^{(1)}, x^{(2)}, \dots, x^{(N)}\}$ 

```

As illustrated in Figure 1, the choice of step size (proposal standard deviation σ_p) impacts convergence and behavior of the sampler. A step size that is too small ($\sigma = 0.1$) causes slow exploration and poor mixing, while an excessively large step ($\sigma = 50.0$) yields a near-zero acceptance rate that stagnates the chain. An optimally tuned step size ($\sigma = 2.0$) balances these extremes, ensuring proper state-space exploration to accurately recover the target density.

While carefully tuning the step size allows RWMH to explore simple, closely-spaced multimodal distributions, the algorithm fails when the modes are widely separated by deep, low-probability valleys. The sampler gets permanently stuck in the mode it discovers first, failing to consider other modes and recover the true distribution. This is because the algorithm proposes jumps local to the current mode, so it cannot cross the wide, low-probability valleys that separate the different modes.

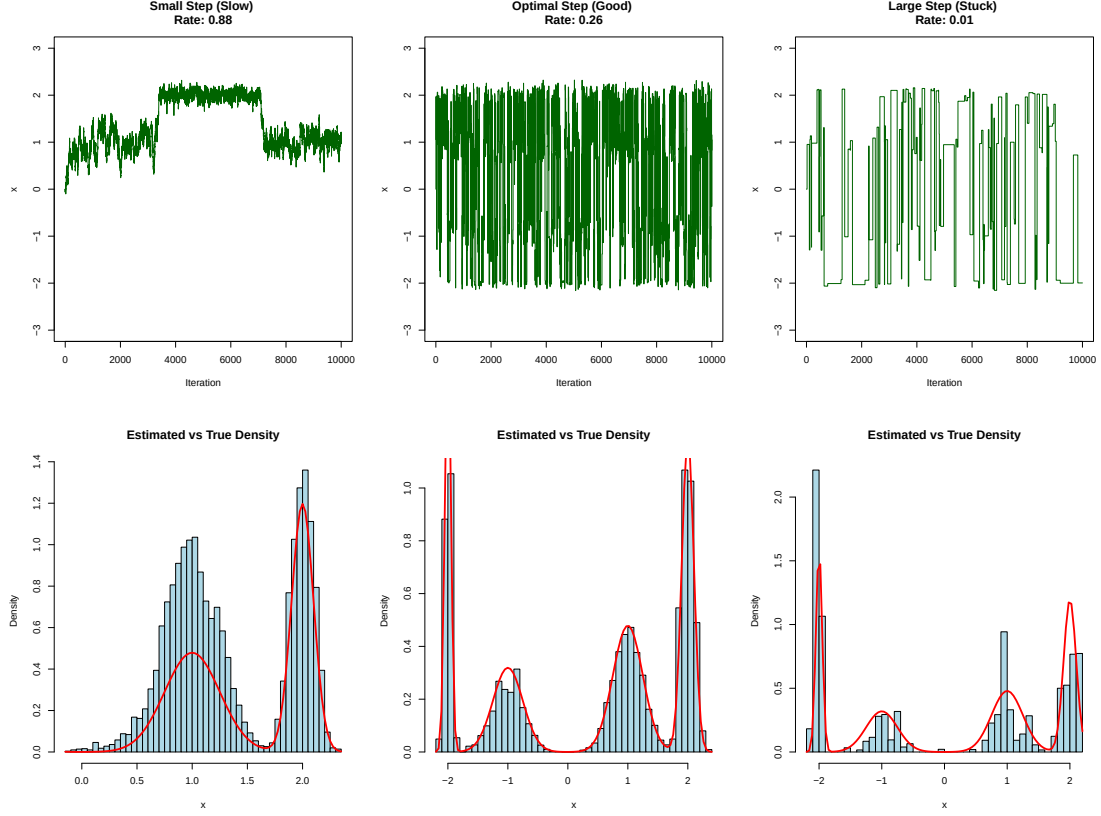


Figure 1: Comparison of trace plots and histograms for Small, Optimal, and Large step sizes for standard Gaussian target.

2 Overcoming Entrapment: Standard Parallel Tempering

To overcome the mode entrapment issues of the basic MH algorithm and properly evaluate barrier-crossing capabilities across the sharp peaks and deep valleys of the previously defined Gaussian mixture, standard Parallel Tempering simulates N Markov chains simultaneously across a temperature ladder $T_1 = 1 < T_2 < \dots < T_N$. Each chain k targets a modified distribution $\pi_k(x) \propto \exp(-U(x)/T_k)$, where higher temperatures flatten the peaks and raise the valleys, allowing standard Random Walk Metropolis updates to cross barriers easily. Periodically, adjacent chains propose state swaps governed by a Metropolis acceptance probability based on the energy difference ΔE and the inverse temperature difference $\Delta(1/T)$. This state-swapping mechanism enables the unflattened target chain ($T_1 = 1$) to successfully pass through low-probability barriers and transition

between fully disconnected modes, as shown in Figure 2.

Algorithm 2 Standard Parallel Tempering (PT)

Require: Potential energy function $U(x) = -\ln \pi(x)$, temperature ladder $1 = T_1 < T_2 < \dots < T_N$, number of iterations M

1: Initialize states $X^{(0)} = \{x_1^{(0)}, x_2^{(0)}, \dots, x_N^{(0)}\}$ for all N chains

2: **for** $i = 0$ to $M - 1$ **do**

3: // **Local Exploration Step**

4: **for** $k = 1$ to N **do**

5: Propose a candidate state $y_k \sim q(y_k | x_k^{(i)})$

6: Calculate local acceptance probability:

$$\alpha_{\text{local}} = \min \left(1, \exp \left(-\frac{(U(y_k) - U(x_k^{(i)}))}{T_k} \right) \frac{q(x_k^{(i)} | y_k)}{q(y_k | x_k^{(i)})} \right)$$

7: Draw $u \sim \text{Uniform}(0, 1)$

8: **if** $u < \alpha_{\text{local}}$ **then**

9: $x_k^{(i+1)} = y_k$

10: **else**

11: $x_k^{(i+1)} = x_k^{(i)}$

12: **end if**

13: **end for**

14: // **Swap Step**

15: **for** $k = 1$ to $N - 1$ **do**

16: Calculate swap acceptance probability for adjacent chains k and $k + 1$:

$$\alpha_{\text{swap}} = \min \left(1, \exp \left(\left(\frac{1}{T_k} - \frac{1}{T_{k+1}} \right) (U(x_k^{(i+1)}) - U(x_{k+1}^{(i+1)})) \right) \right)$$

17: Draw $v \sim \text{Uniform}(0, 1)$

18: **if** $v < \alpha_{\text{swap}}$ **then**

19: Swap the states: $x_k^{(i+1)} \leftrightarrow x_{k+1}^{(i+1)}$

20: **end if**

21: **end for**

22: **end for**

23: **return** Samples $\{x_1^{(1)}, x_1^{(2)}, \dots, x_1^{(M)}\}$ from the target chain ($T_1 = 1$)

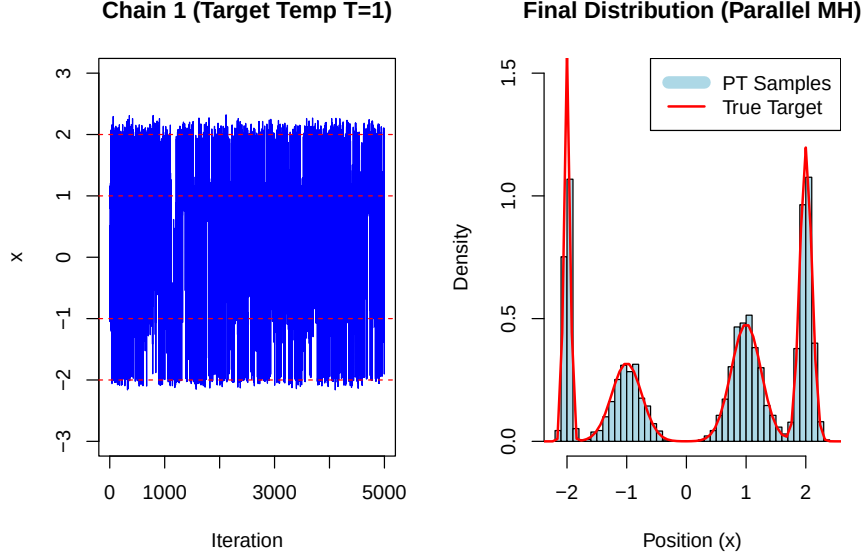


Figure 2: Standard Parallel Tempering with Metropolis Hasting with Mixture of Gaussian Target. The target chain successfully jumps between modes.

2.1 Integrability Limitations of Standard Temperature Scaling

While standard Parallel Tempering (PT) helps in state-space exploration by flattening the distribution, it has a theoretical limitation: for heavy-tailed distributions, high-temperature schedules often yield improper (non-integrable) probability densities.

In standard PT, the tempered density at temperature $T > 1$ is defined as $\pi_T(x) \propto \exp\left(-\frac{U(x)}{T}\right) = (\pi(x))^{\frac{1}{T}}$. Geometrically, raising a probability density to a fractional power $1/T \in (0, 1)$ thickens its tails. If the target density $\pi(x)$ decays rapidly (e.g., the Gaussian distribution), the tempered tails remain integrable for any finite T . However, if the target exhibits polynomial decay, high temperatures will cause the total probability density to diverge.

Example 1. To illustrate this mathematically, consider a standard Cauchy distribution as the target density, where $\pi(x) \propto (1 + x^2)^{-1}$. The corresponding potential energy is $U(x) = \ln(1 + x^2)$. Under standard temperature scaling, the unnormalized tempered density becomes:

$$\pi_T(x) \propto \exp\left(-\frac{\ln(1 + x^2)}{T}\right) = \frac{1}{(1 + x^2)^{\frac{1}{T}}}. \quad (3)$$

For this tempered distribution to represent a proper probability measure, its integral

over $\mathbb{R}$ must be finite: $\int_{-\infty}^{\infty} (1+x^2)^{-1/T} dx < \infty$. By the standard integration limits for polynomial decay, this integral converges only if the tail exponent satisfies $\frac{2}{T} > 1$. Therefore, integrability is limited to temperatures where $T < 2$.

Consequently, if we require a temperature of $T \geq 2$ to traverse, the standard PT algorithm fails. The high-temperature chain evaluates an improper density with infinite mass, causing the Markov chain to diverge toward infinity and destroying the swap mechanism.

3 Moreau-Yosida Envelopes

The Moreau envelope $M_U^t(x)$ serves as a smoothed, continuously differentiable approximation of the target potential $U(x) = -\ln(\pi(x))$, defined mathematically for a smoothing parameter $t > 0$ as:

$$M_U^t(x) = \inf_{y \in \mathbb{R}^d} \left(U(y) + \frac{1}{2t} \|x - y\|^2 \right). \quad (4)$$

This formulation guarantees that the envelope acts as a global lower bound to the original function, $M_U^t(x) \leq U(x)$, and converges pointwise to the original potential as the smoothing parameter approaches zero ($M_U^t(x) \rightarrow U(x)$ as $t \rightarrow 0$) [1], motivating its use as an approximation of potential energy.

For the potential U , the proximal operator, which identifies the exact point in the parameter space that achieves this infimum, is defined as:

$$\text{prox}_{tU}(x) = \operatorname{argmin}_{y \in \mathbb{R}^d} \left(U(y) + \frac{1}{2t} \|x - y\|^2 \right). \quad (5)$$

3.1 Effect of Smoothing Parameter t

The impact of the smoothing parameter t is visualized by plotting the potential energy $M_U^t(x)$ and its corresponding probability density $\pi^t(x) \propto \exp(-M_U^t(x))$ across a sequence of values. To ensure numerical consistency and avoid local minima traps during evaluation, a discrete grid-search optimization is utilized. The potential $U(x)$ is calculated over a master grid $\mathcal{G}$, and the envelope value $M^t(x)$ is computed by minimizing the objective $U(y) + \frac{1}{2t}(x - y)^2$ over this grid space.

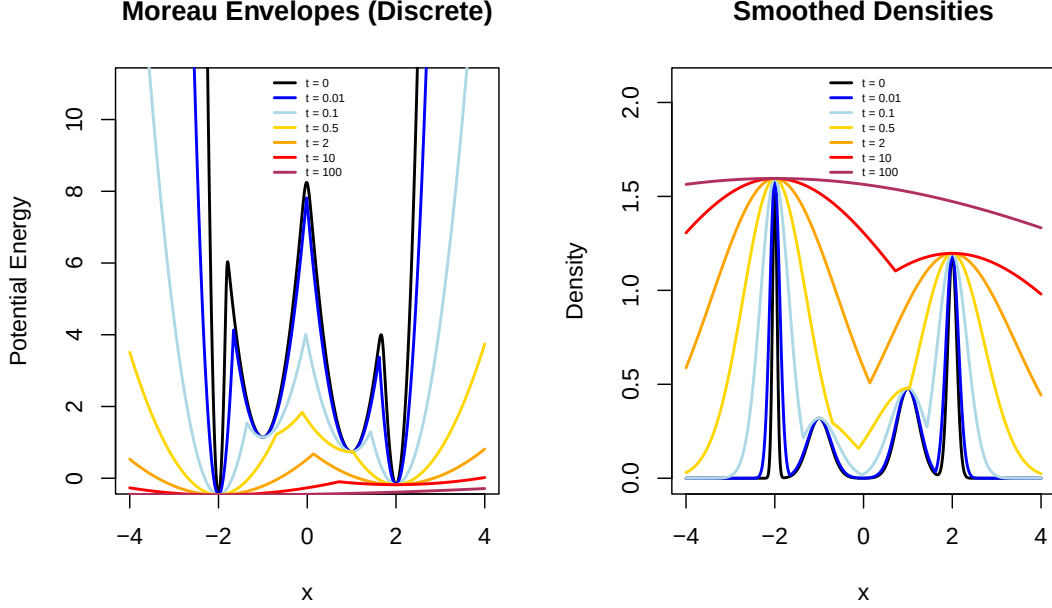


Figure 3: Left: The Potential Energy plot. Right: The corresponding Probability Density. The black line ($t = 0$) represents the highly multimodal target, while the red/maroon lines ($t \geq 5$) illustrate the convexified, unimodal approximations.

Figure 3 demonstrate the geometric transformation induced by the Moreau envelope. At $t = 0$, the target potential exhibits severe multimodality with four distinct local minima. As the smoothing parameter t increases, these separated local minima are gradually smoothed and merged into a single, fully convex shape with one global minimum. Consequently, the corresponding probability density broadens into a wide, unimodal distribution. This convexification allows a Markov chain at higher t values to freely traverse the entire domain and cross energy barriers before progressively annealing back to the true target distribution.

3.2 Convexification of the Potential

To justify the visual smoothing observed in our plots, we can formally measure how the Moreau envelope reduces the non-convexity of a target distribution, established by Habring et al. [1] in their analysis of Moreau-enveloped potentials.

Definition 1 (Nonconvexity). The nonconvexity of a function $H : \mathbb{R}^d \rightarrow \mathbb{R}$ is defined

as the supremum of its deviation from convexity:

$$NC(H) := \sup_{\substack{x, y \in \mathbb{R}^d \\ \lambda \in [0, 1]}} [H(\lambda x + (1 - \lambda)y) - \lambda H(x) - (1 - \lambda)H(y)]. \quad (6)$$

Note: If H is fully convex, $NC(H) = 0$. A larger $NC(H)$ indicates a landscape with more severe local minima and energy barriers.

Lemma 1. *Assuming the proximal operator $\text{prox}_{tH}(x)$ is nonempty for all x , the non-convexity of the Moreau envelope M_H^t is bounded by the nonconvexity of the original function H :*

$$NC(M_H^t) \leq NC(H). \quad (7)$$

Proof Sketch. Let x and x' be arbitrary points, and let p and p' be their respective proximal points. We evaluate the envelope at a convex combination $x_\lambda = \lambda x + (1 - \lambda)x'$.

Using the definition of the Moreau envelope as an infimum, we can bound the non-convexity equation for M_H^t by expanding it and separating the terms into two distinct components:

1. The Original Function Component: This groups the H terms into $H(p_\lambda) - \lambda H(p) - (1 - \lambda)H(p')$. By the definition of nonconvexity, this block is strictly bounded by $NC(H)$.
2. The Distance Penalty Component: This groups the remaining squared L_2 norm terms $(\frac{1}{2t}\|\cdot\|^2)$. Because the squared Euclidean distance is a strictly convex function, its corresponding nonconvexity evaluation is guaranteed to be ≤ 0 .

Since the distance penalty component is always negative or zero, the sum of these two components can never exceed $NC(H)$. Taking the supremum over all points concludes that $NC(M_H^t) \leq NC(H)$. $\square$

3.3 Analytical Derivation of the Convexification Parameter ρ

For a non-convex potential energy function $U(x) = -\ln \pi(x)$, we seek a semiconvexity parameter $\rho > 0$ such that adding a quadratic penalty $\frac{\rho}{2}\|x\|^2$ makes the modified landscape strictly convex. This parameter ρ corresponds to the maximum negative curvature (maximum value of second derivative) of $U(x)$.

3.3.1 Univariate Gaussian Mixture Model

Consider a one-dimensional Gaussian mixture model comprising K components, each defined by a weight w_i , a mean μ_i , and a variance σ_i^2 . The probability density function is given by:

$$\pi(x) = \sum_{i=1}^K w_i \phi(x|\mu_i, \sigma_i^2) = \sum_{i=1}^K \frac{w_i}{\sqrt{2\pi\sigma_i^2}} \exp\left(-\frac{(x-\mu_i)^2}{2\sigma_i^2}\right), \quad (8)$$

where $\phi(x|\mu, \sigma^2)$ denotes the probability density function of a normal distribution with mean μ and standard deviation σ . To simplify the derivatives, we first define the relative contribution of component i to the total density at point x , denoted as $\gamma_i(x)$. This sums to 1 ($\sum_{i=1}^K \gamma_i(x) = 1$):

$$\gamma_i(x) = \frac{w_i \phi(x|\mu_i, \sigma_i^2)}{\pi(x)}. \quad (9)$$

The potential energy is $U(x) = -\ln \pi(x)$. Using the chain rule, the first derivative of the potential is $U'(x) = -\frac{\pi'(x)}{\pi(x)}$. We evaluate the first derivative of the density $\pi'(x)$:

$$\pi'(x) = \sum_{i=1}^K w_i \phi'(x|\mu_i, \sigma_i^2) = \sum_{i=1}^K w_i \left(-\frac{x-\mu_i}{\sigma_i^2}\right) \phi(x|\mu_i, \sigma_i^2). \quad (10)$$

Dividing this by $\pi(x)$ and substituting our definition of $\gamma_i(x)$, the first derivative of the potential simplifies to:

$$U'(x) = \sum_{i=1}^K \gamma_i(x) \left(\frac{x-\mu_i}{\sigma_i^2}\right). \quad (11)$$

Next, we calculate the second derivative. By the quotient rule, $U''(x) = -\frac{\pi''(x)}{\pi(x)} + \left(\frac{\pi'(x)}{\pi(x)}\right)^2$. First, we find the second derivative of the density $\pi''(x)$:

$$\pi''(x) = \sum_{i=1}^K w_i \left(\frac{(x-\mu_i)^2}{\sigma_i^4} - \frac{1}{\sigma_i^2}\right) \phi(x|\mu_i, \sigma_i^2). \quad (12)$$

Dividing by $\pi(x)$ and combining the terms, the second derivative of the potential energy is:

$$U''(x) = \sum_{i=1}^K \gamma_i(x) \left(\frac{1}{\sigma_i^2} - \frac{(x-\mu_i)^2}{\sigma_i^4}\right) + \left(\sum_{i=1}^K \gamma_i(x) \frac{x-\mu_i}{\sigma_i^2}\right)^2. \quad (13)$$

To ensure the convexity of the potential, the second derivative must be non-negative globally: $U''(x) + \rho \geq 0$ for all x . Therefore, the parameter ρ must be strictly greater than maximum value of $-U''(x)$:

$$\rho = \sup_x (-U''(x)). \quad (14)$$

Substituting the derivation, the convexification parameter ρ for any arbitrary Gaussian mixture is formally defined as:

$$\rho = \sup_x \left[\sum_{i=1}^K \gamma_i(x) \left(\frac{(x - \mu_i)^2}{\sigma_i^4} - \frac{1}{\sigma_i^2} \right) - \left(\sum_{i=1}^K \gamma_i(x) \frac{x - \mu_i}{\sigma_i^2} \right)^2 \right]. \quad (15)$$

Finding a closed-form algebraic solution for the global supremum in Equation (15) is intractable for $K > 2$, ρ is determined computationally. This supremum is evaluated by applying a numerical optimization algorithm initialized near these saddle points, which corresponds to the local maxima of the potential separating the individual Gaussian peaks. It effectively targets the localized regions where the maximum negative curvature is guaranteed to occur.

3.3.2 Multivariate Gaussian Mixture Model

This logic extends to high-dimensional parameter spaces. Consider a generic d -dimensional Gaussian mixture model comprising K components, where each component is defined by a weight w_i , a mean vector $\mu_i \in \mathbb{R}^d$, and a covariance matrix $\Sigma_i \in \mathbb{R}^{d \times d}$. The probability density function is:

$$\pi(x) = \sum_{i=1}^K w_i \phi(x|\mu_i, \Sigma_i), \quad (16)$$

where $\phi(x|\mu, \Sigma)$ denotes the probability density function of a normal distribution of d -dimensional with mean μ and standard deviation Σ . Just as in the univariate case, we define $\gamma_i(x)$ for the i -th component at a given point x :

$$\gamma_i(x) = \frac{w_i \phi(x|\mu_i, \Sigma_i)}{\pi(x)}. \quad (17)$$

For the potential energy $U(x) = -\ln \pi(x)$, the first derivative becomes a gradient vector $\nabla U(x) = -\frac{\nabla \pi(x)}{\pi(x)}$. Evaluating the gradient of the multivariate Gaussian density and

substituting the $\gamma_i(x)$ function yields:

$$\nabla U(x) = \sum_{i=1}^K \gamma_i(x) \Sigma_i^{-1} (x - \mu_i). \quad (18)$$

To measure curvature in a multidimensional space, we must calculate the Hessian matrix $\nabla^2 U(x)$. Applying the multivariate quotient rule, the Hessian evaluates to:

$$\nabla^2 U(x) = -\frac{\nabla^2 \pi(x)}{\pi(x)} + \frac{\nabla \pi(x) \nabla \pi(x)^T}{\pi(x)^2}. \quad (19)$$

Expanding this using the derivatives of the individual Gaussian components, the Hessian matrix simplifies into:

$$\begin{aligned} \nabla^2 U(x) = & \sum_{i=1}^K \gamma_i(x) (\Sigma_i^{-1} - \Sigma_i^{-1} (x - \mu_i) (x - \mu_i)^T \Sigma_i^{-1}) \\ & + \left(\sum_{i=1}^K \gamma_i(x) \Sigma_i^{-1} (x - \mu_i) \right) \left(\sum_{i=1}^K \gamma_i(x) \Sigma_i^{-1} (x - \mu_i) \right)^T. \end{aligned} \quad (20)$$

To ensure the multidimensional landscape is completely convexified, the modified Hessian must be positive semi-definite globally: $\nabla^2 U(x) + \rho I \succeq 0$. Therefore, the parameter ρ must be strictly greater than the most negative curvature across all possible directions in the parameter space. Mathematically, ρ corresponds to the magnitude of the most negative eigenvalue of the Hessian matrix:

$$\rho = \sup_x \lambda_{\max} \left(-\nabla^2 U(x) \right). \quad (21)$$

Since calculating an exact algebraic maximum for complex, high-dimensional mixtures is practically impossible, we simply evaluate the Hessian's eigenvalues at the saddle points between the peaks.

3.3.3 Relationship Between ρ and Smoothing Parameter t

The derivation of the maximum negative curvature ρ establishes a theoretical limit on the Moreau-Yosida smoothing parameter t . For the Moreau envelope to be well-defined, continuously differentiable, and computationally stable for gradient-based samplers, the objective function of the infimum must be strictly convex. Recall the definition of the

Moreau envelope for the potential energy $U(x)$:

$$M_U^t(x) = \inf_y \left\{ U(y) + \frac{1}{2t} \|x - y\|^2 \right\}. \quad (22)$$

Let the function inside the infimum be defined as $h_x(y)$:

$$h_x(y) = U(y) + \frac{1}{2t} \|x - y\|^2. \quad (23)$$

To determine the convexity of $h_x(y)$, we evaluate its Hessian matrix with respect to y . The Hessian of the quadratic penalty term $\frac{1}{2t} \|x - y\|^2$ is $\frac{1}{t} I$, where I is the identity matrix. Therefore, the Hessian of the total objective evaluates to:

$$\nabla^2 h_x(y) = \nabla^2 U(y) + \frac{1}{t} I. \quad (24)$$

For $h_x(y)$ to be strictly convex, its Hessian must be positive definite everywhere, meaning $\nabla^2 h_x(y) \succ 0$.

By our previous derivation, ρ represents the absolute maximum negative curvature of $U(x)$. This means that the most negative eigenvalue of $\nabla^2 U(y)$ is bounded by $-\rho$, yielding the matrix inequality $\nabla^2 U(y) \succeq -\rho I$ for all $y \in \mathbb{R}^d$.

Substituting this condition into the strict convexity requirement gives:

$$-\rho I + \frac{1}{t} I \succ 0. \quad (25)$$

Factoring out the identity matrix separates the scalar inequality:

$$\frac{1}{t} - \rho > 0 \implies t < \frac{1}{\rho}. \quad (26)$$

This relationship defines a threshold for strict convexity of the smoothing parameter t . While $t < 1/\rho$ guarantees that the Moreau envelope is continuously differentiable everywhere, Parallel Tempering relies on large t values to flatten the potential energy. If the smoothing parameter t becomes too large ($t \geq 1/\rho$), the quadratic penalty becomes too weak to overpower the negative curvature ρ of the distribution.

3.4 Theoretical Guarantee: Proper Density of the Smoothed Approximation

To analyze the envelope's properties, the total potential energy is generally decomposed as $U(x) = f(x) + g(x)$, where $f(x)$ is a continuously differentiable, lower-bounded function, and $g(x)$ is a proper, lower semi-continuous convex function representing the non-smooth component [1]. When applying the Moreau envelope to modify the target distribution, it is imperative to guarantee that the resulting smoothed approximation $U^t(x) = f(x) + M_g^t(x)$ remains a valid, integrable probability distribution. [1][6]

This framework naturally covers two algorithmic approaches: enveloping the entire energy landscape (where $f(x) = 0$ and $g(x) = U(x)$), or enveloping only a specific non-smooth penalty (where $f(x)$ is the smooth log-likelihood and $g(x)$ is the penalty).

Proposition 1. *Let $g(x)$ be a proper, lower semi-continuous convex function such that either $e^{-g(x)}$ is integrable with respect to the Lebesgue measure, or $g(x)$ is Lipschitz continuous. Then, for any smoothing parameter $t > 0$, the smoothed approximation $\pi^t(x) \propto e^{-U^t(x)}$ defines a proper probability measure, satisfying:*

$$0 < \int_{\mathbb{R}^d} e^{-U^t(x)} dx < +\infty. \quad (27)$$

Proof. The proof must establish both the strict positivity and the finite upper bound of the integral. By the fundamental definition of the Moreau envelope as an infimum, $M_g^t(x) \leq g(x)$ for all $x \in \mathbb{R}^d$. Consequently, $U^t(x) \leq U(x)$, which immediately provides the strict lower bound: $0 < \int e^{-U(x)} dx < \int e^{-U^t(x)} dx$ [6].

To prove the integral is finitely bounded from above (integrability), the two assumptions regarding $g(x)$ are evaluated independently:

Case 1: $g(x)$ is Lipschitz continuous: If $g(x)$ is Lipschitz, the deviation between the original function and its Moreau envelope is strictly bounded by the squared Lipschitz constant of g , such that $g(x) - M_g^t(x) \leq \frac{t}{2} \|g\|_{Lip}^2$. Exponentiating this relationship yields $e^{-M_g^t(x)} \leq e^{-g(x)} e^{\frac{t}{2} \|g\|_{Lip}^2}$. Because $f(x)$ is assumed to be lower-bounded and $e^{-g(x)}$ is integrable, the strict boundedness of the deviation ensures that $\int e^{-U^t(x)} dx < +\infty$.

Case 2: $e^{-g(x)}$ is integrable: If $g(x)$ is not Lipschitz, its integrability and convexity guarantee that it can be bounded from below by a linear cone function: $g(x) \geq M_1 + \rho_g \|x - x_g\|$ for some constants $M_1 \in \mathbb{R}$ and $\rho_g > 0$. By evaluating the Moreau envelope of this lower bound, it can be shown that $M_g^t(x)$ is similarly bounded below by a smoothed cone function: $M_g^t(x) \geq M_1 + h^t(x)$, where $h^t(x)$ is the envelope of $\rho_g \|x - x_g\|$. Because

the proximal operator associated with this norm behaves as a soft-thresholding operator, the resulting envelope strictly maintains at least linear growth at infinity, yielding $h^t(x) \geq \rho_g \|x - x_g\| + M_2$. Therefore, $M_g^t(x)$ grows at least linearly as $\|x\| \rightarrow \infty$, which mathematically guarantees that $e^{-M_g^t(x)}$ decays exponentially and is finitely integrable. Since $f(x)$ is bounded from below, the complete approximated density $e^{-U^t(x)}$ is necessarily integrable. $\square$

4 The DAZ Algorithm

To leverage the convexifying properties of the Moreau envelope, the Diffusion at Absolute Zero (DAZ) algorithm was proposed in recent literature[1] to sample from non-differentiable and non-convex distributions. DAZ integrates the Moreau envelope with Unadjusted Langevin Algorithm (ULA) dynamics, attempting to guide the sampler toward the target distribution by slowly annealing the smoothing parameter t from a large, highly convex state down to zero. While theoretically sound, empirical evaluations reveal severe practical limitations when this method is applied to highly multimodal target landscapes.

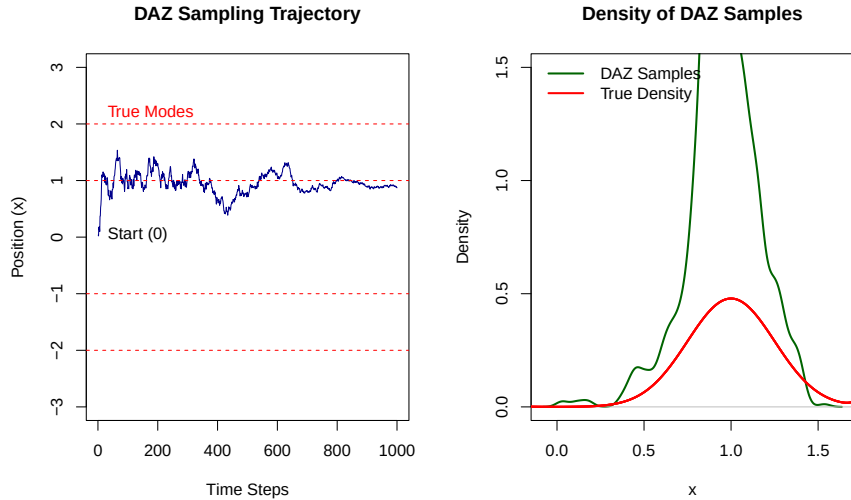


Figure 4: DAZ Failure with Fast Cooling ($N = 50, K = 20$). The sampler becomes permanently trapped in a single mode.

As illustrated in Figure 4, utilizing a standard, fast-cooling annealing schedule causes the DAZ sampler to become immediately trapped in a single local minimum, entirely

failing to explore the broader state space. To counter this entrapment, an aggressively slow cooling schedule with a massive computational budget can be employed (Figure 5). However, while the sampler eventually traverses the energy barriers, the resulting trajectory suffers from poor mixing. The chain remains stuck in individual modes for long periods, yielding highly correlated samples at a prohibitive computational cost. These inefficiencies demonstrate that DAZ alone is insufficient for multiple mode exploration, directly motivating the integration of Moreau envelopes with state-swapping mechanics in Moreau-Yosida Parallel Tempering.

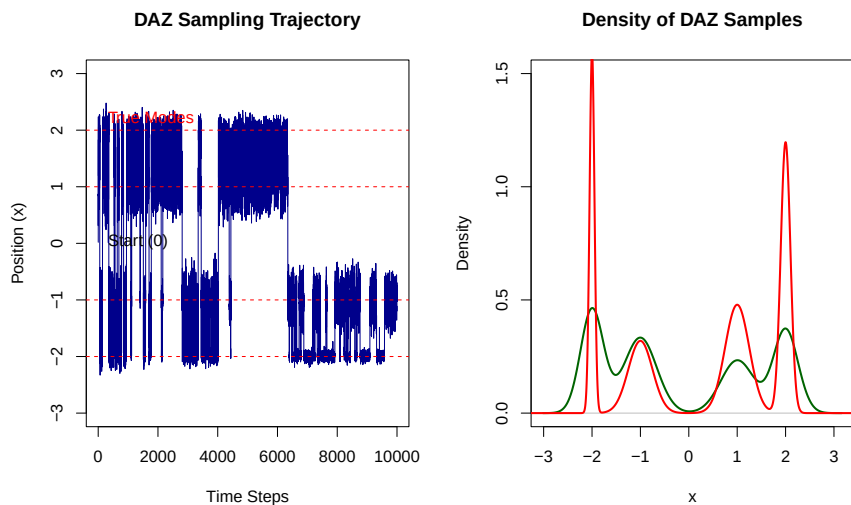


Figure 5: DAZ with Slow Cooling ($N = 5000, K = 2000$). The sampler eventually visits all modes, but exhibits poor mixing and requires an extreme computational cost.

5 Moreau-Yosida Parallel Tempering (MYPT)

To overcome the severe mixing inefficiencies and mode entrapment issue of the DAZ algorithm, the structural convexification of Moreau Envelopes is integrated with the state-swapping mechanics of Parallel Tempering. This yields Moreau-Yosida Parallel Tempering (MYPT). In this framework, the standard temperature ladder is replaced by a smoothing parameter ladder ($t_1 = 0 < t_2 < \dots < t_N$), allowing multiple Markov chains to traverse varying levels of smoothness simultaneously.

5.1 MYPT with Random Walk (MYPT-RW)

The foundational implementation, MYPT-RW, utilizes standard Random Walk Metropolis updates for local chain exploration. However, the intra-chain acceptance ratios and inter-chain swap probabilities are calculated based on the smoothed Moreau potential energy $M^t(x)$. While standard Parallel Tempering merely flattens the target density, MYPT physically convexifies the landscape, completely merging isolated modes at higher levels of t . This structural transformation helps in barrier crossing and robust state swapping before annealing the samples down to the target chain ($t_1 = 0$).

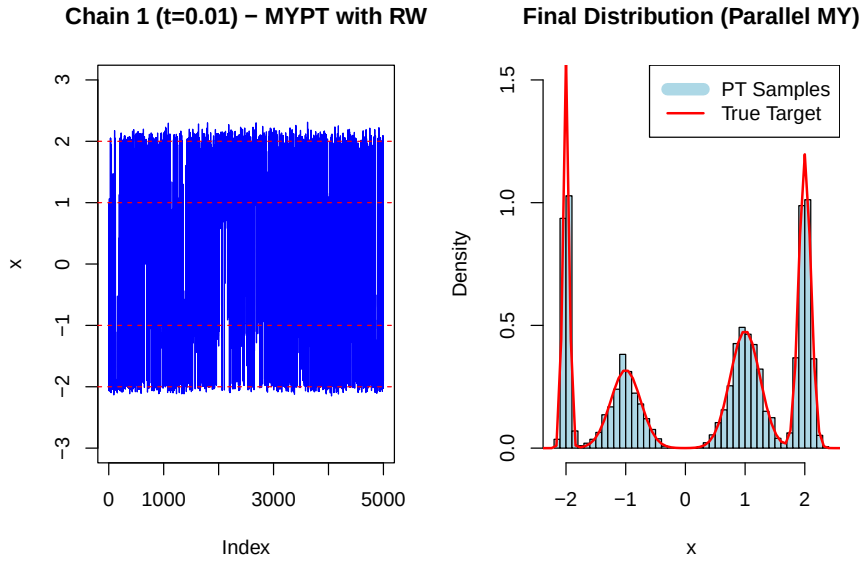


Figure 6: Results for Moreau-Yosida PT utilizing standard Random Walk updates. The chains successfully swap states across the convexified landscapes.

5.2 MYPT with MALA (MYPT-MALA)

To maximize sampling efficiency, the unguided Random Walk is replaced with a Metropolis-Adjusted Langevin Algorithm (MALA) update, yielding MYPT-MALA. This approach utilizes the exact gradient of the envelope to actively guide proposals. Specifically, the proposed state X^* is drawn from a Normal distribution $q(X^*|X_k) \sim \mathcal{N}(X_k - \frac{\tau}{2}\nabla M^{t_k}(X_k), \tau I)$, which is generated via the update rule:

$$X^* = X_k - \frac{\tau}{2}\nabla M^{t_k}(X_k) + \sqrt{\tau}Z, \quad (28)$$

where $Z \sim \mathcal{N}(0, I)$ is standard Gaussian noise, and the gradient of the Moreau envelope is evaluated as $\nabla M^{t_k}(x) = \frac{1}{t_k}(x - \text{prox}_{t_k U}(x))$.

The primary advantage of this method is its ability to utilize the geometry of the smoothed potential to actively pull the sampler toward high-probability regions, while the Metropolis correction step maintains detailed balance. By combining precise gradient guidance with state-swapping across mathematically convexified landscapes, MYPT-MALA effectively solves both non-differentiability and multi-modality entrapment problems simultaneously.

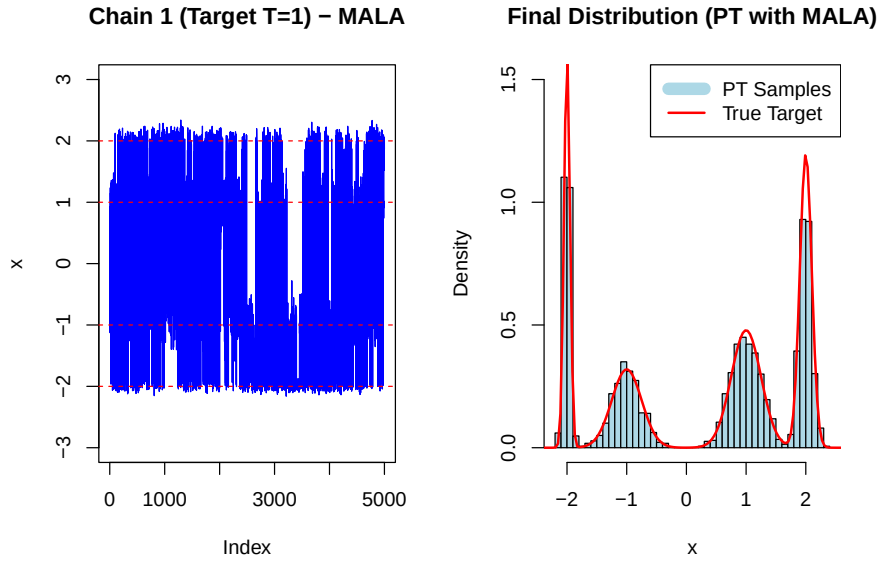


Figure 7: Results for Moreau-Yosida PT utilizing gradient-guided MALA updates, demonstrating highly efficient mode exploration and rapid convergence.

6 Applications and Examples

To evaluate the developed methodologies against complex statistical and spatial problems, the algorithms are applied to two highly challenging target distributions: the non-differentiable Bayesian Lasso and a severely multimodal 20-component Gaussian mixture.

6.1 Evaluating Mode Discovery: 20-Mode Bivariate Gaussian Mixture

To evaluate mode-hopping capabilities on a complex distribution, an asymmetric, bivariate Gaussian mixture comprising 20 distinct modes is constructed. These modes are randomly scattered across a 2D domain with varying covariance structures, correlations, and component weights, heavily separated by expansive regions of near-zero probability.

To explicitly demonstrate the structural transformation induced by the algorithm, the smoothed density $\pi^t(x, y) \propto \exp(-M_U^t(x, y))$ is visualized across the parameter space for the defined smoothing parameter ladder ($t_1 = 0 < t_2 < \dots < t_N$).

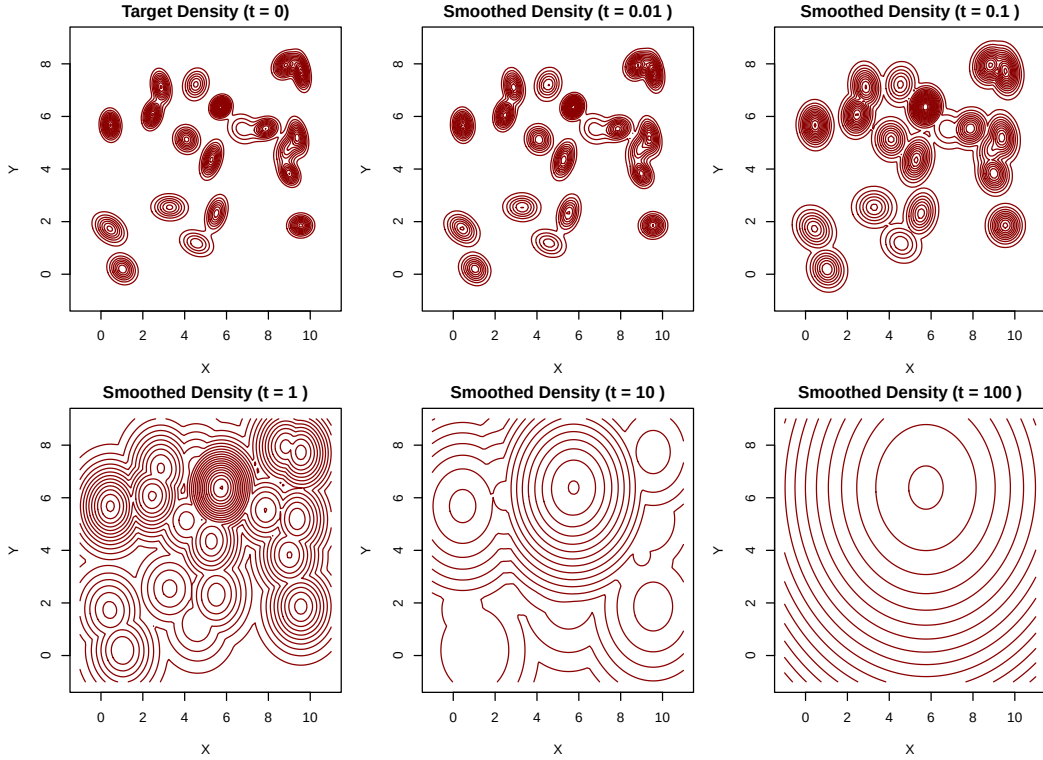


Figure 8: Progression of the smoothed Moreau density contours across the smoothing parameter ladder. The highly multimodal target ($t = 0$) is systematically convexified into a unified basin at higher t values.

As illustrated in Figure 8, the target density at $t = 0$ exhibits severe multimodality with completely isolated peaks. As the smoothing parameter t increases, the Moreau envelope merges these disconnected modes into a single, easily traversable basin. This physical convexification provides the theoretical guarantee that chains operating at higher t levels can freely explore the global domain.

Applying both Standard Parallel Tempering (PT-RW) and Moreau-Yosida Parallel Tempering (MYPT-RW) to this target density demonstrates that both algorithms successfully overcome mode entrapment. Visually, the scatter plots indicate that both methods effectively traverse the low-probability regions to discover and sample from all 20 distinct modes.

However, the underlying mixing dynamics of the Markov chains differ significantly. To quantitatively evaluate and compare this performance, the Expected Squared Jump Distance (ESJD) is calculated for 100 independent replications for the target chains of both algorithms. The results were averaged to provide a highly stable metric for the efficiency of state-space exploration. ESJD measures the average squared Euclidean distance between consecutive states in the Markov chain. A higher ESJD indicates that the sampler takes larger, more effective steps between states rather than exhibiting highly correlated, localized random walk behavior within a single mode.

As summarized in Table 1, the MYPT-RW algorithm achieves a substantially higher average ESJD as compared to the standard PT-RW baseline. This stark quantitative result - nearly doubling the average jump distance - conclusively demonstrates that while standard Parallel Tempering is capable of eventual mode discovery, integrating mathematical convexification via Moreau envelopes facilitates superior transition efficiency and chain mixing across highly rugged, disconnected landscapes.

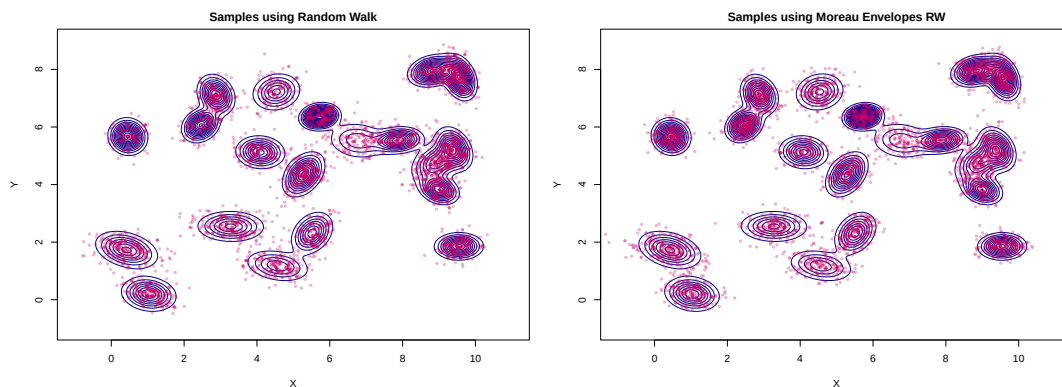


Figure 9: Left: Mode exploration using Standard PT-RW. Right: Mode exploration using MYPT-RW.

Algorithm	Average ESJD (100 Replications)
Standard PT-RW	17.584
MYPT-RW	28.449

Table 1: Comparison of the average Expected Squared Jump Distance (ESJD) over 100 independent replications, demonstrating the enhanced transition efficiency of the Moreau-Yosida smoothed sampler.

6.2 The Bayesian Lasso

The Bayesian Lasso utilizes an unconditional Laplace prior on regression coefficients (β) and an independent, scale-invariant prior on the variance (σ^2), which is equivalent to applying an L_1 penalty in the posterior.

$$\pi(\beta) = \frac{\lambda}{2} e^{-\lambda|\beta|} \quad (29)$$

$$\pi(\sigma^2) \propto \frac{1}{\sigma^2} \quad (30)$$

Combining these priors with a standard Gaussian likelihood yields an unnormalized joint posterior distribution:

$$\pi(\beta, \sigma^2 | \tilde{y}) \propto \left(\frac{1}{\sigma^2} \right)^{\frac{n-1}{2}+1} \exp \left\{ -\frac{1}{2\sigma^2} (\tilde{y} - X\beta)^T (\tilde{y} - X\beta) - \lambda|\beta| \right\}. \quad (31)$$

Inspired by the bimodal posterior structure presented in Appendix B of Park and Casella (2008) [5], the summary statistics are modified to amplify the separation between the modes. The target distribution is parameterized with $n = 10$, $p = 1$, $X^T X = 1$, $X^T \tilde{y} = 8$, $\tilde{y}^T \tilde{y} = 65$, and a penalty parameter $\lambda = 2.4$. Visualizing the transformed joint density $\pi(\beta, \ln(\sigma^2))$ confirms the presence of two distinct, widely separated modes: a standard least squares estimate ($\beta = 8, \ln(\sigma^2) \approx -2$) and a sparse solution ($\beta = 0, \ln(\sigma^2) \approx 2$) (Figure 10).

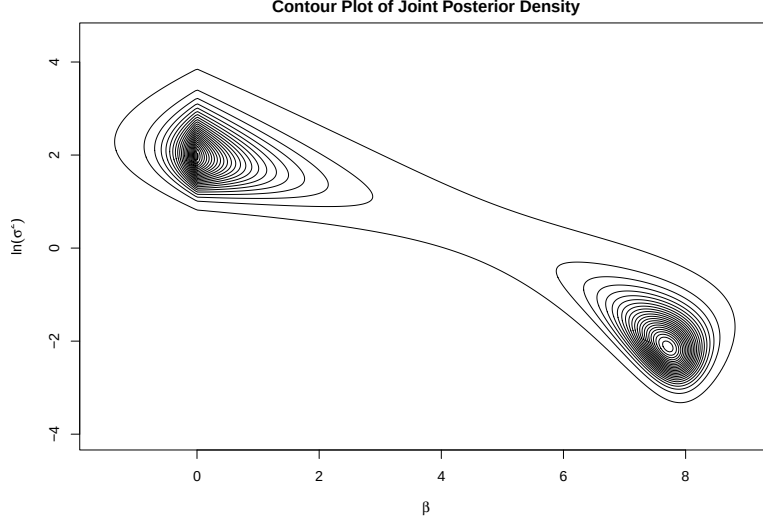


Figure 10: Contour plot of the joint posterior density $\pi(\beta, \ln(\sigma^2))$, exhibiting severe bimodality and a non-differentiable crease at $\beta = 0$.

To sample from this complex distribution, a bivariate Random Walk Metropolis-Hastings algorithm is initially employed within a standard Parallel Tempering framework. While temperature scaling sufficiently flattens low-dimensional energy barriers to allow standard random walks to cross between modes, it cannot resolve the underlying non-differentiability of the L_1 penalty. To explicitly address this non-differentiable crease, Moreau-Yosida Parallel Tempering (MYPT-RW) is applied.

6.2.1 Numerical Moreau Envelope on the Joint Posterior

First, the MYPT-RW algorithm applies the Moreau envelope numerically over the entire potential energy function of the transformed joint posterior. To compute this smoothed energy at any given point, a 2D discrete grid-search over the (β, ν) space is utilized.

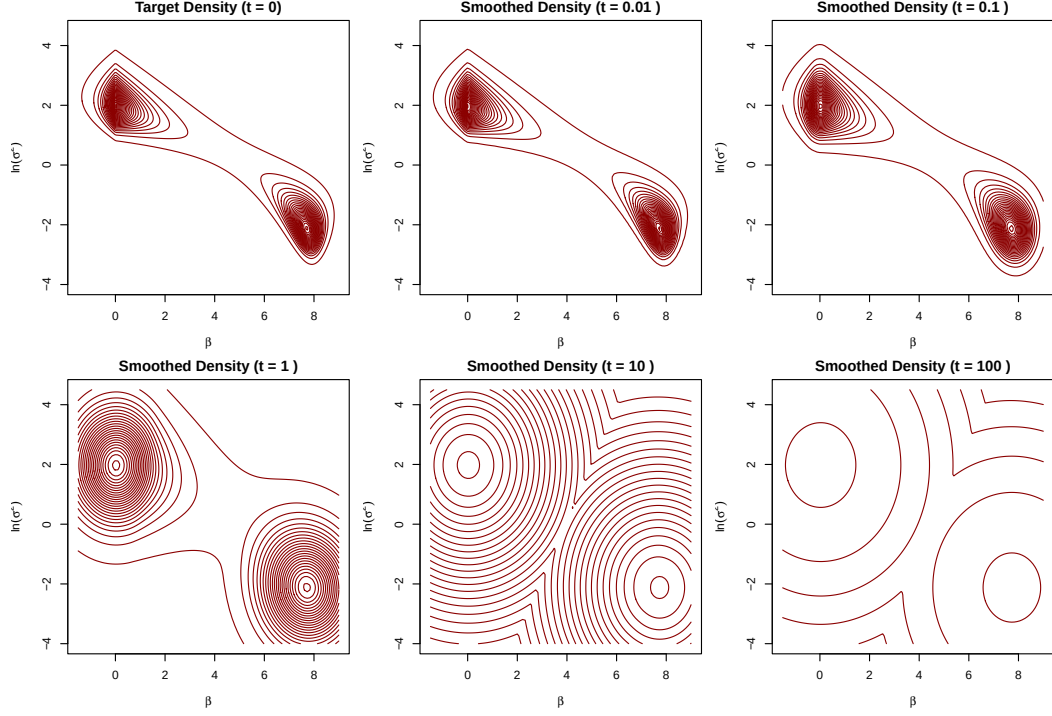


Figure 11: Progression of smoothed density contours when enveloping the entire potential energy function. Both modes are systematically merged into a unified basin at higher t values.

As shown in Figure 11, this approach physically convexifies the entire energy landscape. As the smoothing parameter t increases, the sharp L_1 penalty crease at $\beta = 0$ is smoothed out, and the two isolated peaks are mathematically merged into a single, easily traversable density. However, evaluating a dense 2D grid at every single MCMC iteration is computationally prohibitive, scaling with an $O(N^2)$ algorithmic complexity.

6.2.2 Optimization: Analytical Moreau Envelope for the L_1 Penalty

To drastically improve computational efficiency, the specific structure of the Bayesian Lasso posterior is exploited. Instead of computing the Moreau envelope numerically for the entire joint posterior, the total potential energy $U(\beta, \nu) = -\ln \pi(\beta, \nu | \tilde{y})$ is decomposed into a smooth, continuously differentiable component $f(\beta, \nu)$ and a non-smooth component $g(\beta)$:

$$U(\beta, \nu) = f(\beta, \nu) + g(\beta). \quad (32)$$

The function $f(\beta, \nu)$ represents the smooth portion containing the negative log-likelihood and the variance prior:

$$f(\beta, \nu) = \frac{n-1}{2}\nu + \frac{1}{2e^\nu}(\tilde{y} - X\beta)^T(\tilde{y} - X\beta). \quad (33)$$

The function $g(\beta)$ represents the non-smooth L_1 penalty responsible for the sharp crease:

$$g(\beta) = \lambda|\beta|. \quad (34)$$

By applying the Moreau envelope operator exclusively to the non-smooth penalty $g(\beta)$, the algorithm seeks to solve the following minimization problem:

$$M_g^t(\beta) = \inf_u \left(\lambda|u| + \frac{1}{2t}(\beta - u)^2 \right). \quad (35)$$

Because $g(\beta)$ is the absolute value function scaled by λ , this formulation is equivalent to the proximal operator of the L_1 norm. The exact analytical solution to this minimization evaluates to the well-known Huber function:

$$M_g^t(\beta) = \begin{cases} \frac{\beta^2}{2t} & \text{if } |\beta| \leq t\lambda \\ \lambda|\beta| - \frac{t\lambda^2}{2} & \text{if } |\beta| > t\lambda \end{cases} \quad (36)$$

This piecewise function behaves exactly like the original L_1 penalty when β is far from zero but seamlessly transitions into a differentiable quadratic bowl in the neighborhood of the non-differentiable crease $([-t\lambda, t\lambda])$. The exact, smoothed total potential energy evaluated by the sampler becomes:

$$U^t(\beta, \nu) = f(\beta, \nu) + M_g^t(\beta). \quad (37)$$

Implementing this analytical formulation reduces the algorithmic complexity of evaluating the Moreau energy from a 2D grid search to a single $O(1)$ scalar calculation.

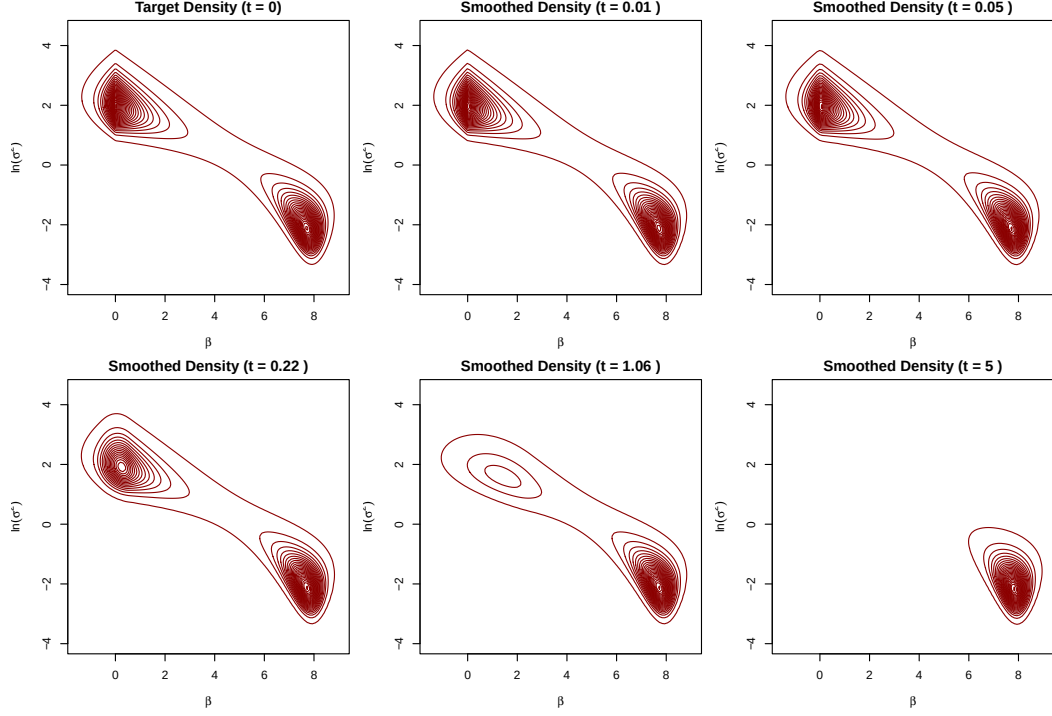


Figure 12: Progression of smoothed density contours when applying the analytical envelope strictly to the L_1 penalty. The sparse mode dissipates entirely at higher t values, leaving only the unpenalized likelihood mode.

As shown in Figure 12, applying the envelope strictly to the L_1 penalty provides a massive algorithmic advantage by causing the sparse mode ($\beta = 0$) to disappear at higher t values. This sparse mode exists entirely because of the sharp, non-differentiable crease of the absolute value function. By flattening this sharp crease into a smooth, quadratic bowl, the physical trap holding the probability mass at zero is erased, and the distribution simplifies into the single least squares mode. This allows the high- t Markov chains to easily escape the sparse trap and freely explore the parameter space. As the states from these unconstrained chains are swapped down the t -ladder toward the target distribution ($t = 0$), the sharp L_1 penalty reconstructs itself around them. This creates a continuous flow of state exchanges that bridges the wide valley, allowing the sampler to perfectly sample from the true bimodal distribution.

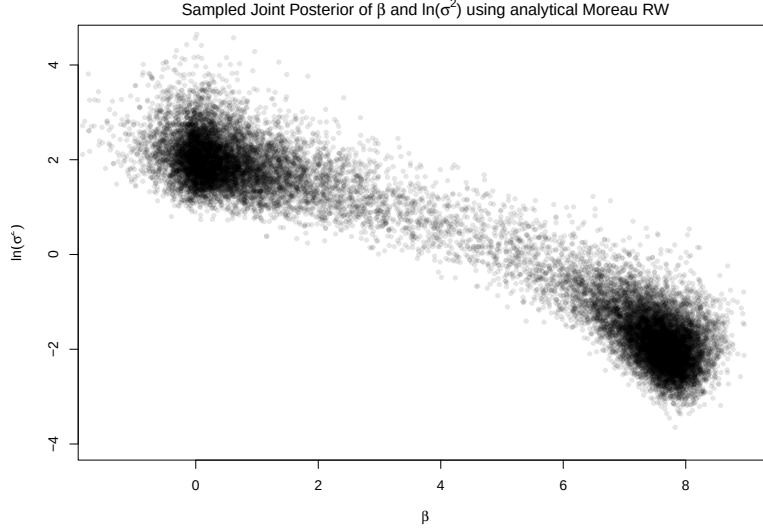


Figure 13: Scatter plot of the sampled joint posterior utilizing the fast, analytical Moreau envelope applied strictly to the L_1 penalty. The sampler successfully bridges the wide, low-probability valley to capture both the sparse mode and the least squares estimate.

6.2.3 Quantitative Performance Comparison

To quantitatively evaluate the mixing efficiency of these state-space exploration strategies, the Expected Squared Jump Distance (ESJD) is calculated and averaged over 100 independent replications for the target chains ($T = 1$ or $t = 0$) of all three implementations. As summarized in Table 2, evaluating the generated trajectories yields an average ESJD of 17.837 for the standard PT-RW algorithm. The numerical MYPT-RW algorithm, which envelopes the entire potential energy function, improves this transition efficiency to 29.851. Finally, the optimized analytical MYPT-RW algorithm, which envelopes strictly the non-differentiable L_1 penalty, achieves the highest overall average ESJD of 32.259. This result confirms that analytically targeting the specific source of non-differentiability not only drastically reduces computational complexity but also increases the efficiency of the algorithm.

Algorithm	Average ESJD (100 Replications)
Standard PT-RW	17.837
Numerical MYPT-RW	29.851
Analytical MYPT-RW	32.259

Table 2: Comparison of the average Expected Squared Jump Distance (ESJD) over 100 independent replications, demonstrating the superior transition efficiency of the analytical Moreau-Yosida approach.

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